PROJECT REPORT ON
AUTO TICKET CLASSIFICATION USING FLOW DESIGNER

Submitted by

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1. INTRODUCTION

1.1 Project Overview

The school IT helpdesk receives multiple incident requests from students and teachers, including Wi-Fi issues, projector failures, password problems and slow computers. The supplied project reference states that IT staff manually review each request and assign a category, making the process time-consuming and inefficient.

The proposed system automates ticket classification using ServiceNow Flow Designer by analysing keywords in the Short Description and Description fields. It automatically assigns the appropriate Category and Subcategory and sends an acknowledgement email to the caller.

1.2 Purpose

The purpose is to reduce manual effort, improve ticket-routing efficiency, standardize ticket information and provide a maintainable no-code automation solution.

2. IDEATION PHASE

2.1 Problem Statement

The helpdesk process requires staff to identify the issue type, select the correct category and subcategory, and inform the caller about ticket creation. The reference identifies this as time consuming, error-prone and difficult to scale.

2.2 Empathy Map Canvas

Aspect	Project Interpretation
Says	Student or teacher reports an IT problem.
Thinks	Expects the request to be recorded and routed correctly.
Does	Creates a ticket with caller information and issue description.
Feels	Needs simple submission and timely confirmation.
Pain Points	Manual categorization and possible classification errors.
Needs	Automatic classification, structured data and notification.

2.3 Brainstorming

- Keyword-based ticket classification
- Automatic Category and Subcategory assignment
- Dependent Category/Subcategory choices
- Record-created Flow Designer trigger
- Automated caller email
- No-code and maintainable implementation
- Future extension with more categories and automation

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	System Action	Outcome
Issue occurs	Identifies IT problem	—	Support required
Ticket creation	Enters caller and description	Creates record and auto number	Ticket recorded
Classification	Does not manually choose category	Flow checks keywords	Category/Subcategory assigned
Notification	Waits for confirmation	Sends email	Submission acknowledged
Support	IT staff handles ticket	Uses structured data	Efficient routing

3.2 Solution Requirement

- Automatic classification from issue description
- Automatic Category and Subcategory
- Dependent choice logic
- Automated email notification
- Structured ticket storage
- Easy maintenance and future scalability

3.3 Data Flow Diagram

Student / Teacher → Incident Workflow Record → Record Created Trigger → Keyword Conditions → Category & Subcategory Update → Email Notification → IT Helpdesk

3.4 Technology Stack

Component	Purpose
ServiceNow	IT service management platform
Flow Designer	No-code workflow automation
Incident Workflow table	Ticket record storage
Update Set	Configuration packaging and transfer
Email action	Caller notification
Reference fields	User and group relationships

4. PROJECT DESIGN

4.1 Problem Solution Fit

The design replaces repeated manual categorization with a predefined Flow Designer workflow. The documented conditions cover Wi-Fi, projector, password/login and slow/hanging computer issues.

4.2 Proposed Solution

A custom Incident WorkFlow table stores Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned to. Category, Subcategory and State are choice fields; caller, group and assigned user are reference fields.
The documented dependency mapping is: Wi-Fi → Network; Projector → Hardware; Forgot Password → Access; Slow Computer → Performance.

4.3 Solution Architecture

User → Incident Workflow Form → Record Created → Category Empty Check → If/Else If Keyword Logic → Update Record → Send Email → IT Helpdesk

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activities	Deliverable
Ideation	Problem identification and brainstorming	Project concept

Requirement Analysis	Business requirements and update set	Requirements
Backend Configuration	Table and fields	Configured data model
Automation	Trigger, conditions, update actions, email	Working flow
Testing	Wi-Fi, projector and email checks	Test results
Deployment	Complete update set and XML export	Deployable update set
Documentation	Final report and demonstration	Project report

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The supplied reference documents functional validation and data checks but does not provide numerical response-time or throughput measurements. Therefore, this report does not add unsupported performance figures.

- Mandatory fields captured correctly
- Auto-number generated without duplication
- Category and Subcategory stored accurately
- Reference fields resolve correctly

Test Case 1 – Wi-Fi Issue

Item	Documented Result
Short Description	WiFi not working in library
Category	Network
Subcategory	WiFi
Email	Sent to caller

Test Case 2 – Projector Issue

Item	Documented Result
Short Description	Projector not turning on
Category	Hardware
Subcategory	Projector

Email	Sent to caller

Email Verification

The reference instructs the tester to open System Logs → Emails, search for the configured subject, open the email record and preview the mail.

7. RESULTS

7.1 Output Screenshots

The final report should contain screenshots from the actual ServiceNow implementation. The supplied PDF describes the following output/configuration views:

1. Incident WorkFlow table
2. Configured fields
3. Category/Subcategory dependency
4. Flow Designer trigger
5. Wi-Fi classification
6. Projector classification
7. Password classification
8. Slow Computer classification
9. Email action
10. Wi-Fi test result
11. Projector test result
12. Email log/preview
13. Completed Update Set and XML export

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Reduces manual classification effort
- Improves consistency
- No-code Flow Designer implementation
- Dependent choice logic
- Automatic email acknowledgement
- Maintainable and scalable design

8.2 Disadvantages / Limitations

- Keyword-based classification depends on predefined conditions
- Documented flow covers only configured issue patterns
- New issue types require new conditions
- Reference does not document machine-learning classification
- No numerical performance benchmark is provided

9. CONCLUSION

The Auto Ticket Classification using Flow Designer project provides an end-to-end automation solution for a school IT helpdesk. It automates ticket classification from short-description keywords, reduces manual effort, standardizes ticket data and sends an acknowledgement email.

The implementation combines a structured Incident WorkFlow table, dependent Category/Subcategory choices, Flow Designer automation and email notification. Testing in the supplied reference demonstrates automatic classification for Wi-Fi and projector scenarios.

The reference identifies future possibilities including assignment automation, SLA tracking and predictive intelligence. These are future enhancements rather than implemented features claimed in the source.

APPENDIX – IMPLEMENTATION STEPS

A. Update Set Creation

1. All menus → Update Sets → Local Update Sets
2. Click New
3. Name: Project Update Set
4. State: In progress
5. Application: Global
6. Save and Make This My Current Set

B. Custom Table and Fields

1. Create Incident WorkFlow table
2. Enable Auto Number
3. Configure the required fields
4. Add choices to Category, Subcategory and State

C. Dependency

1. Configure Dictionary for Subcategory
2. Open Advanced view
3. Enable Use dependent field
4. Select Category as Dependent on field
5. Configure the four documented mappings

D. Flow Designer

1. Create Auto Classify School IT Tickets
2. Trigger: Record Created
3. Table: Incident Workflow
4. Condition: Category is Empty
5. Add If/Else If conditions
6. Update Category/Subcategory
7. Send email to Caller → Email
8. Activate the flow

E. Deployment

1. Open Project Update Set
2. Change In progress to Complete
3. Save
4. Export to XML